

$$\text{Number of seats sold} = \frac{2x}{3}. \text{ Number of seats unsold} = \left(x - \frac{2x}{3}\right) = \frac{x}{3}.$$

$$\text{Number of sold seats in lower deck} = \frac{4}{5} \text{ of } \frac{x}{3} = \frac{x}{5}.$$

$$\text{Number of unsold seats in lower deck} = \left(\frac{x}{3} - \frac{x}{5}\right) = \frac{x}{15}.$$

$$\therefore \text{ Required fraction} = \frac{x/15}{x/3} = \frac{1}{5}.$$

Ex. 37. In a certain office, $\frac{1}{3}$ of the workers are women, $\frac{1}{2}$ of the women are married and $\frac{1}{3}$ of the married women have children. If $\frac{3}{4}$ of the men are married and $\frac{2}{3}$ of the married men have children, what part of workers are without children?

Sol. Let the total number of workers be x . Then,

$$\text{Number of women} = \frac{x}{3} \text{ and number of men} = \left(x - \frac{x}{3}\right) = \frac{2x}{3}.$$

$$\text{Number of women having children} = \frac{1}{3} \text{ of } \frac{1}{2} \text{ of } \frac{x}{3} = \frac{x}{18}.$$

$$\text{Number of men having children} = \frac{2}{3} \text{ of } \frac{3}{4} \text{ of } \frac{2x}{3} = \frac{x}{3}.$$

$$\text{Number of workers having children} = \left(\frac{x}{18} + \frac{x}{3}\right) = \frac{7x}{18}.$$

$$\therefore \text{ Workers having no children} = \left(x - \frac{7x}{18}\right) = \frac{11x}{18} = \frac{11}{18} \text{ of all workers.}$$

Ex. 38. A crate of mangoes contains one bruised mango for every 30 mangoes in the crate. If 3 out of every 4 bruised mangoes are considered unsaleable, and there are 12 unsaleable mangoes in the crate, then how many mangoes are there in the crate?

Sol. Let the total number of mangoes in the crate be x . Then,

$$\text{Number of bruised mangoes} = \frac{1}{30}x.$$

$$\text{Number of unsaleable mangoes} = \left(\frac{3}{4} \times \frac{1}{30}x\right) = \frac{1}{40}x.$$

$$\therefore \frac{1}{40}x = 12 \text{ or } x = (12 \times 40) = 480.$$

Hence, total number of mangoes in the crate = 480.

Ex. 39. The cost of 4 rings and 2 bangles is ₹ 57200. What is the cost of 6 rings and 3 bangles? (Bank Recruitment, 2009)

$$\text{Sol. We have: } 4R + 2B = 57200 \Rightarrow \frac{3}{2}(4R + 2B) = \frac{3}{2} \times 57200 \Rightarrow 6R + 3B = 85800.$$

Hence, cost of 6 rings and 3 bangles = ₹ 85800.

Ex. 40. The cost of 15 kg of sugar is ₹ 255, the cost of 17 kg of tea is ₹ 1615 and the cost of 22 kg of rice is ₹ 572. What is the total cost of 18 kg of sugar, 21 kg of tea and 27 kg of rice? (Bank P.O., 2008)

$$\text{Sol. Cost of 1 kg sugar} = ₹ \left(\frac{255}{15}\right) = ₹ 17; \text{ Cost of 1 kg tea} = ₹ \left(\frac{1615}{17}\right) = ₹ 95;$$

$$\text{Cost of 1 kg rice} = ₹ \left(\frac{572}{22}\right) = ₹ 26.$$

$$\therefore \text{ Required cost} = ₹ (17 \times 18 + 95 \times 21 + 26 \times 27) = ₹ (306 + 1995 + 702) = ₹ 3003.$$